In this test, you have to prepare 3 **specifications** with 3-3 **test cases**, the **reduction tables** and small **algorithms** using the specification tool and the structogram tool. Prepare the solutions based on the patterns.
In the postcondition (of the specification), you should use the short form - and the algorithm needs to follow your specification.
Please look after the **good, readable quality** of your work.
There are **90 minutes** available. For a good solution **a maximum of 100 points** could be reached. Don't forget your name and Neptun-code! During work you are not allowed to talk (chat), to use cell phones, the internet, books, tutorials, ...
Remember! Cheating is not allowed, and action will be taken.
Create a pdf: File/Export/Create a pdf -> **Upload in the TMS**

Name: <write your name> **Neptun code:** <your neptun code>

1st task

Marta takes part in a spelling bee. The words she spelled were stored with the information on whether he spelled the word correctly or not. Which words did she spell correctly?

example input:
n: 5
words:

word	*correct*
bired	false
canyou	false
card	true
wter	false
car	true

Output: [card, car]

Specification

<copy the specification>

Link of the specification:
<copy the link of your specification – use the "Share" button>

Reduction table:
<prepare the reduction table>

Test cases:
<Write your test cases (3) in the specification tool. The link of the spec.tool includes then it – but please, copy here, too>

1st test case	2nd test case	3rd test case

Algorithm

<insert the picture(s) of your algorithm>

Link of the algorithm:
<copy the link of your algorithm – use the "Share" button>

2nd task

Trurl recorded the concentration of black dust at “m” location in the Great Shroud Desert for “n” days. Give a location where the concentration was increasing every day.

Example input:
n: 5, m: 3

Output: 2

	1.loc.	2.loc.	3.loc.
1.day	12	2	0
2.day	4	5	1
3.day	1	10	2
4.day	0	12	3
5.day	1	14	1

Specification

<copy the specification>

Link of the specification:
<copy the link of your specification – use the "Share" button>

Reduction table:
<prepare the reduction table>

Test cases:
<Write your test cases (3) in the specification tool. The link of the spec.tool includes then it – but please, copy here, too>

1st test case	2nd test case	3rd test case

Algorithm

<insert the picture(s) of your algorithm>

Link of the algorithm:
<copy the link of your algorithm – use the "Share" button>

3rd task

Pepi the Pirate kept a record of where his boat had docked. Was there a port where he had docked more than k times?

Example input:
n: 10, k: 3
ports: [Shanghai, Singapore, Cape Town, Singapore, Honduras, Singapore, Honduras, Shanghai, Singapore, Cape Town]

Output: yes

Specification

<copy the specification>

Link of the specification:
<copy the link of your specification – use the "Share" button>

Reduction table:
<prepare the reduction table>

Test cases:
<Write your test cases (3) in the specification tool. The link of the spec.tool includes then it – but please, copy here, too>

1st test case	2nd test case	3rd test case

Algorithm

<insert the picture(s) of your algorithm>

Link of the algorithm:
<copy the link of your algorithm – use the "Share" button>